

- 4 (a) Define *electric potential*.

.....
 [1]

- (b) Two point charges P and Q are separated by a horizontal distance of 30 mm as shown in Fig. 4.1. P has a charge of $+20\ \mu\text{C}$ and Q has a charge of $-40\ \mu\text{C}$.

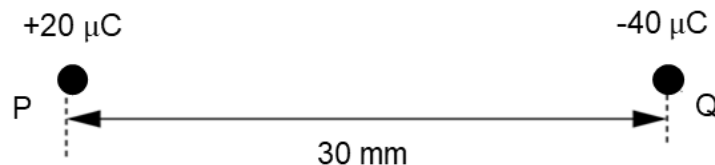


Fig 4.1

- (i) The horizontal distance from P towards Q is r . On Fig. 4.2, sketch the variation with r of the electric potential V due to both point charges P and Q.

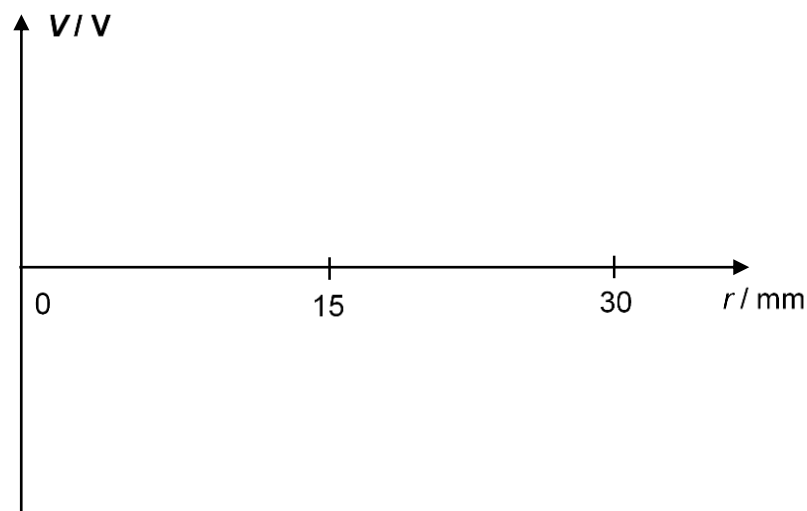


Fig 4.2

[2]

- (ii) On Fig. 4.3, draw electric field lines to represent the electric field in the region between P and Q for the charge configuration.



Fig 4.3

[3]

- (iii) Determine the electric potential energy of Q.

electric potential energy = J [2]

[Total: 8]